

V3 Pipeline — Egitim Raporu

Tarih: 2026-03-22 21:14

PIPELINE OZELLIKLERI:

Feature Engineering: V3 (Label-encoded predictions + ayri OHE + manuel k-mer)
Feature sayisi: 979
Toplam ornek: 4287 (CV: 3429, Hold-Out: 858)
Model turleri: LightGBM, XGBoost
Hyperparameter: Grid Search (12 combo/model, 3-fold CV)
Degerlendirme: 5-Fold Stratified CV + %20 Hold-Out Test

5-FOLD CV SONUCLARI (egitim seti %80):

lightgbm : F1 = 0.9846 +/- 0.0011
xgboost : F1 = 0.9846 +/- 0.0009

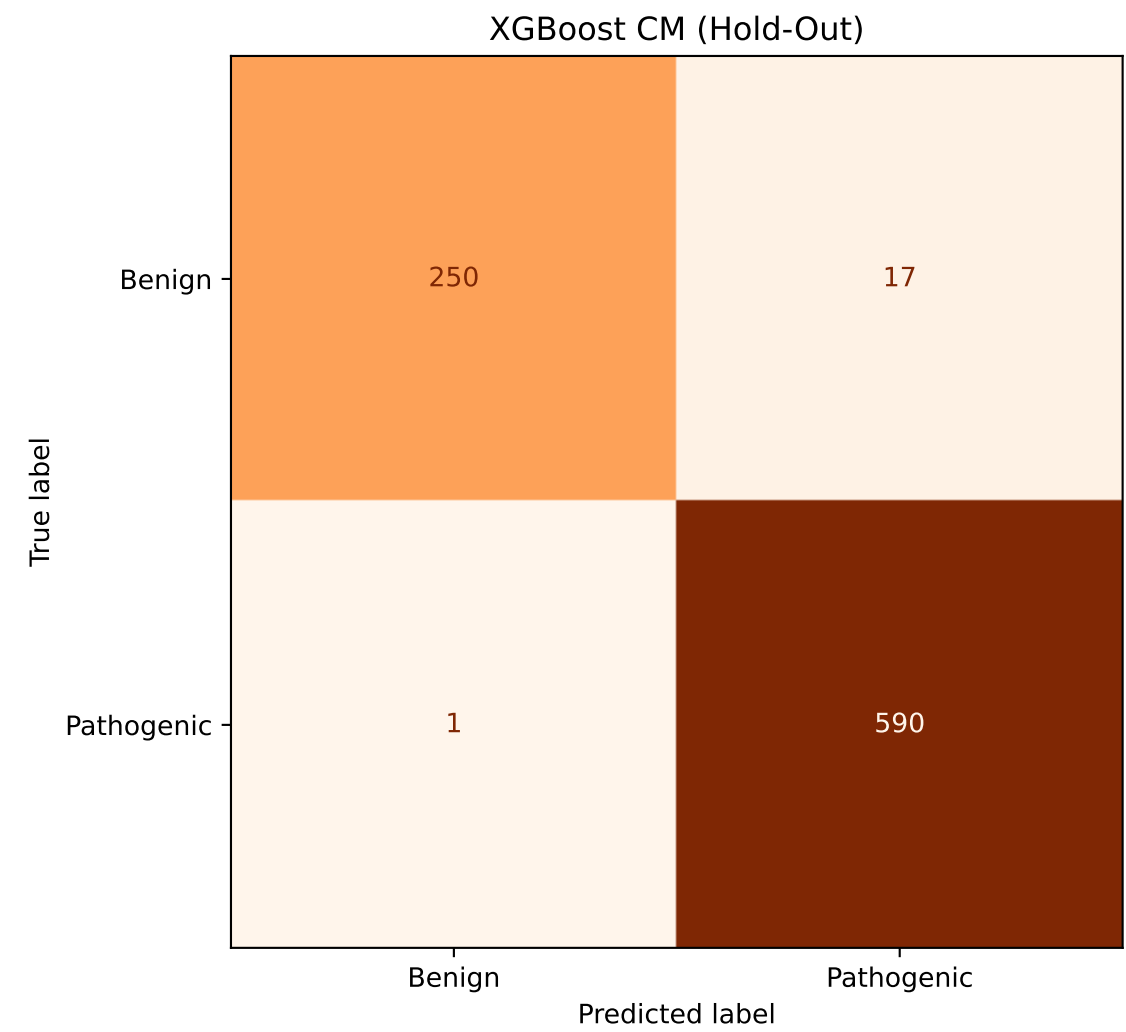
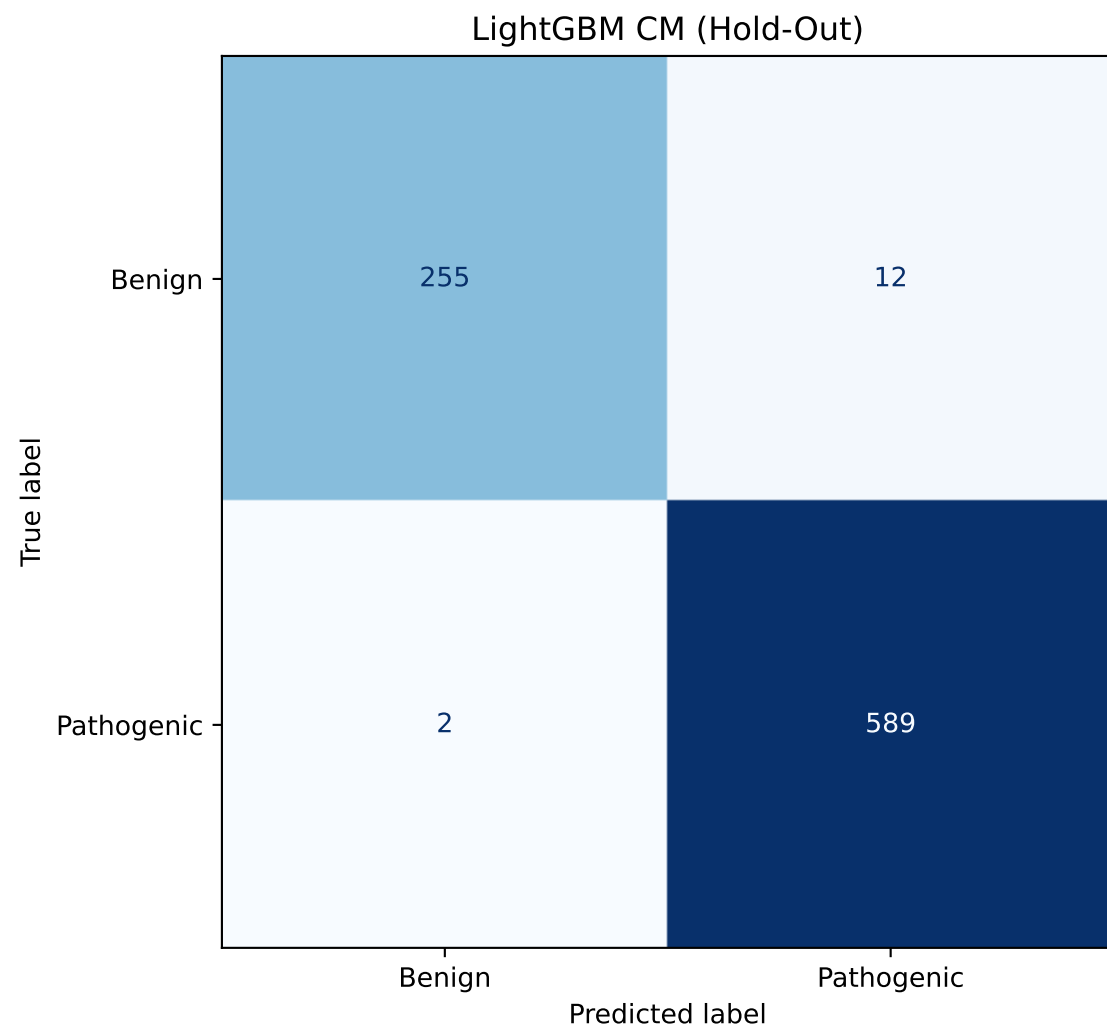
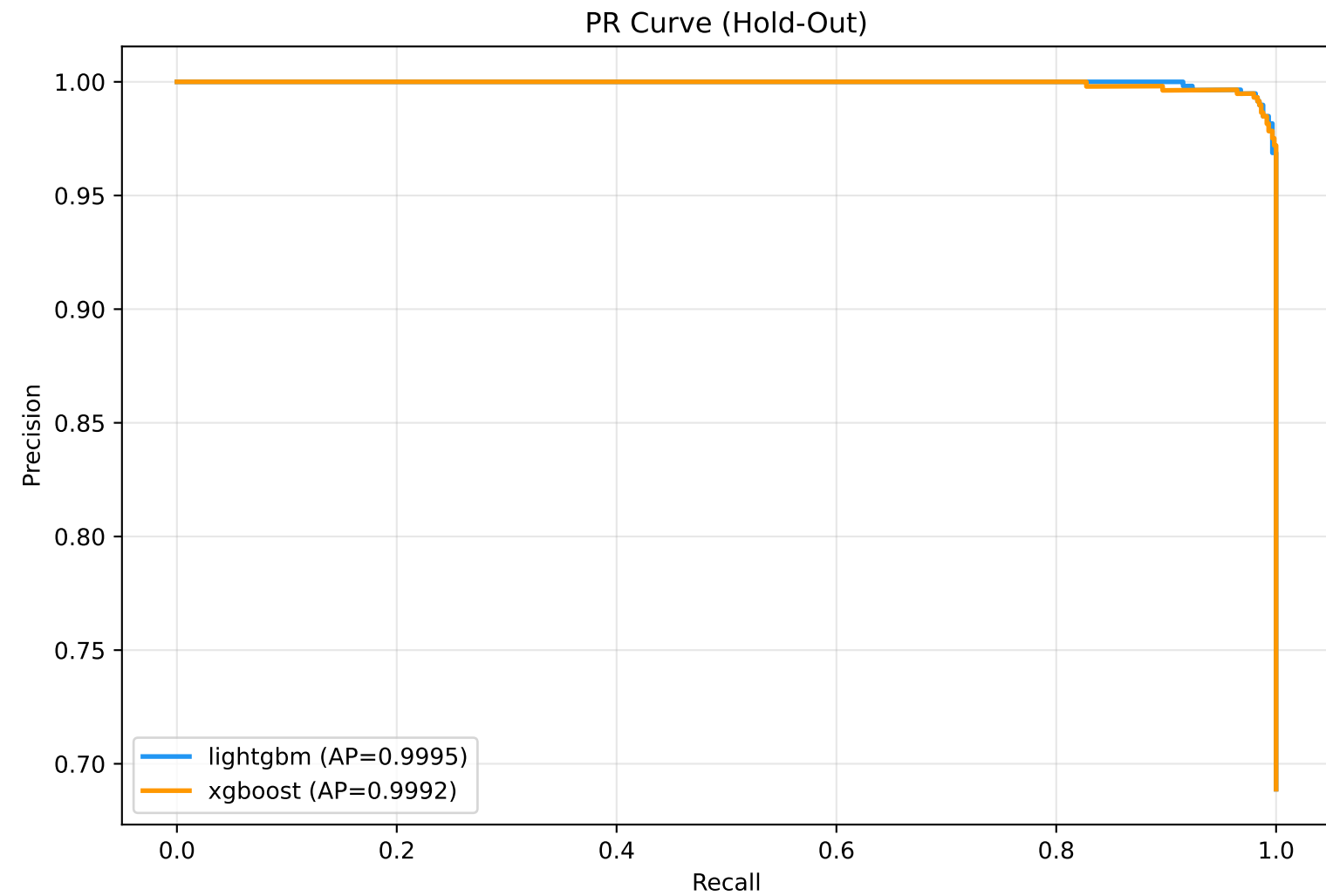
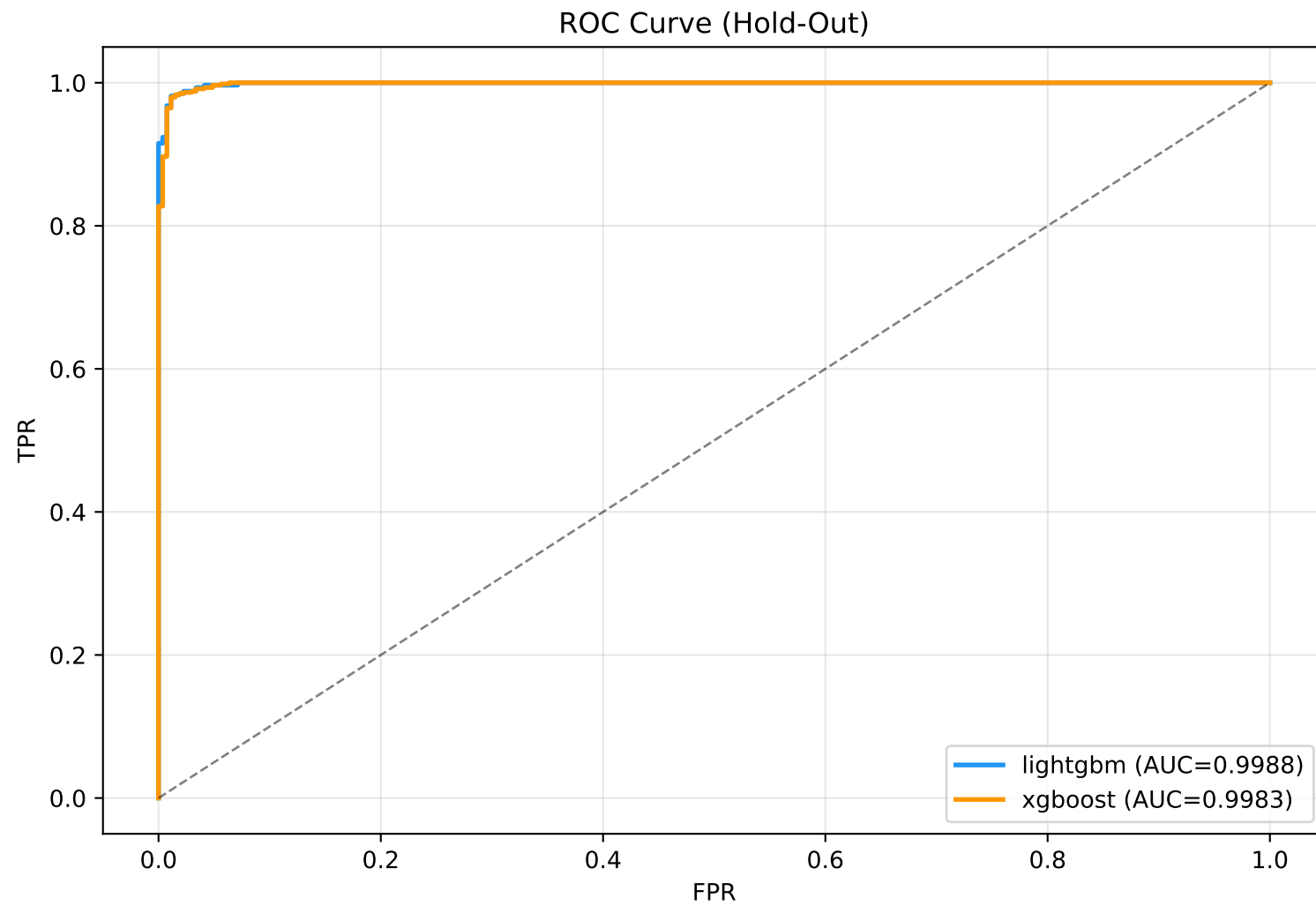
HOLD-OUT TEST SONUCLARI (hic gorulmemis %20):

lightgbm : F1=0.9883 AUC=0.9988 (CV F1=0.9846)
xgboost : F1=0.9850 AUC=0.9983 (CV F1=0.9846)

KARSILASTIRMA (full_fe vs v3 hold-out):

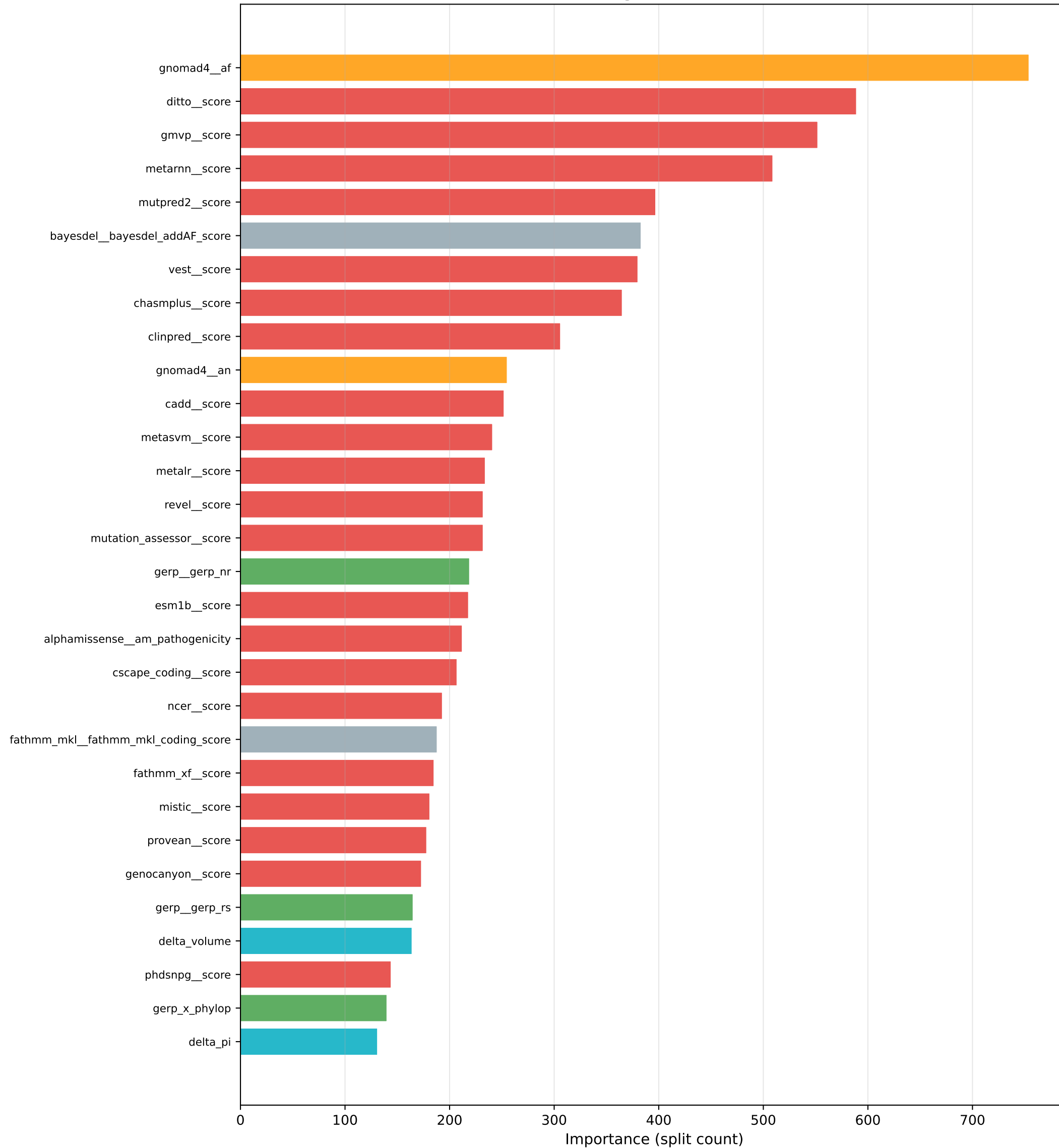
lightgbm: full_fe F1=0.6967 -> v3 Hold-Out F1=0.9883 (+0.2916)
xgboost: full_fe F1=0.6879 -> v3 Hold-Out F1=0.9850 (+0.2971)

Hold-Out Test Sonuclari (%20 — Hic Gorulmemis Veri)

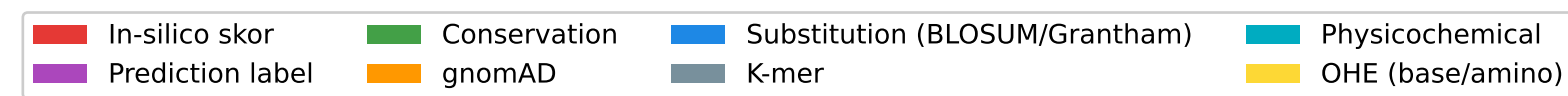
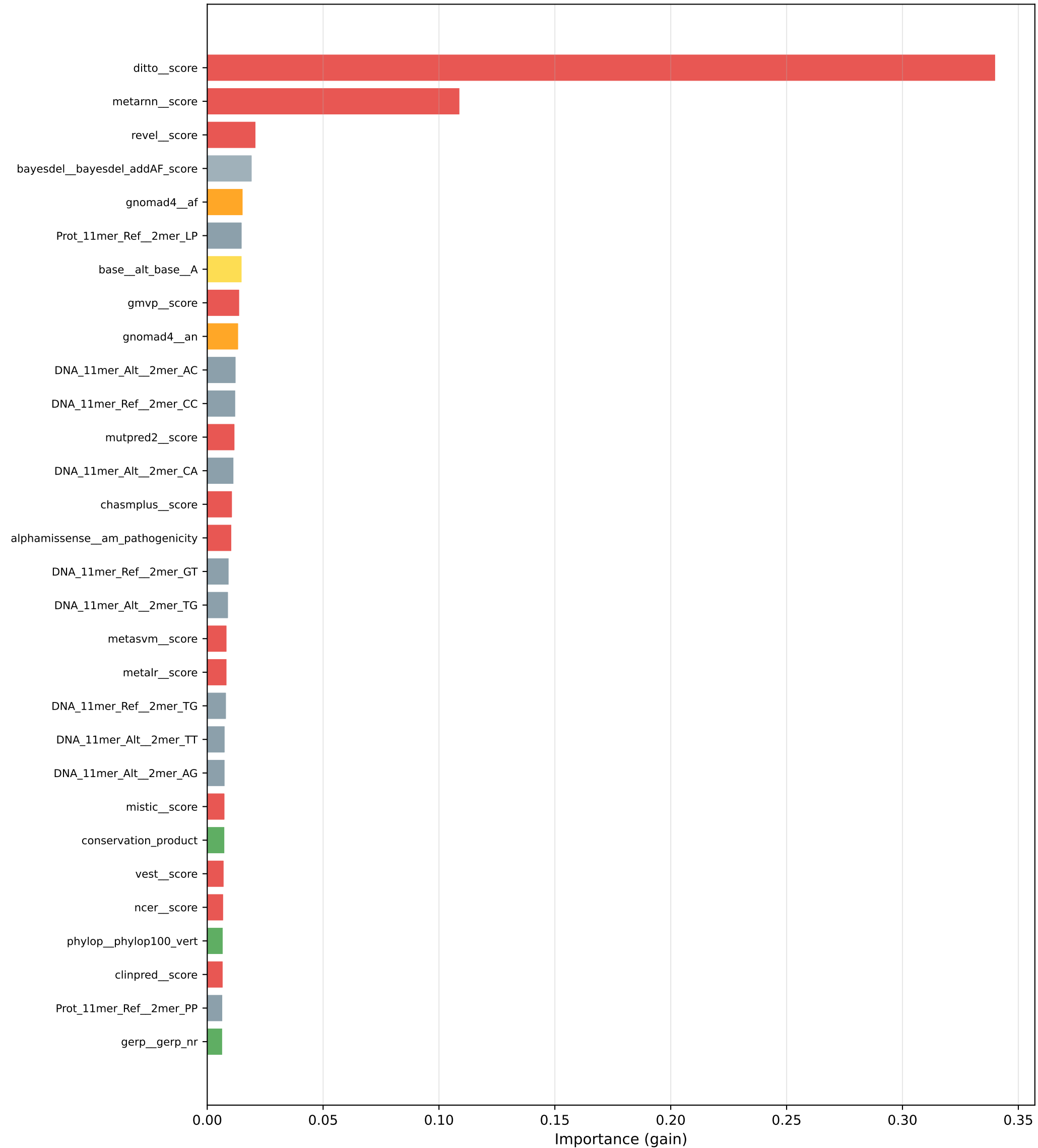


Feature Importance — Top 30 (En İyi CV Fold Modeli)

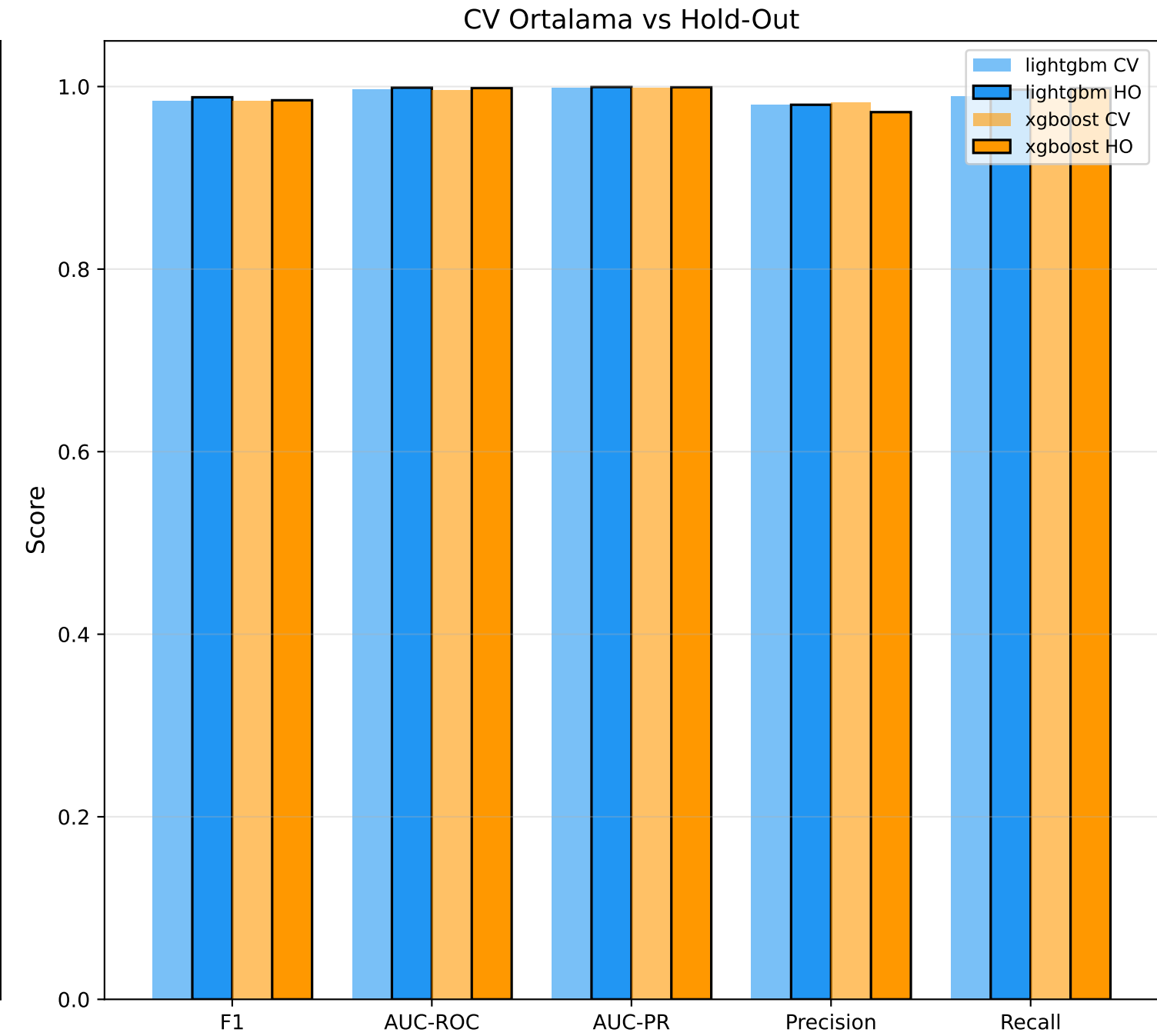
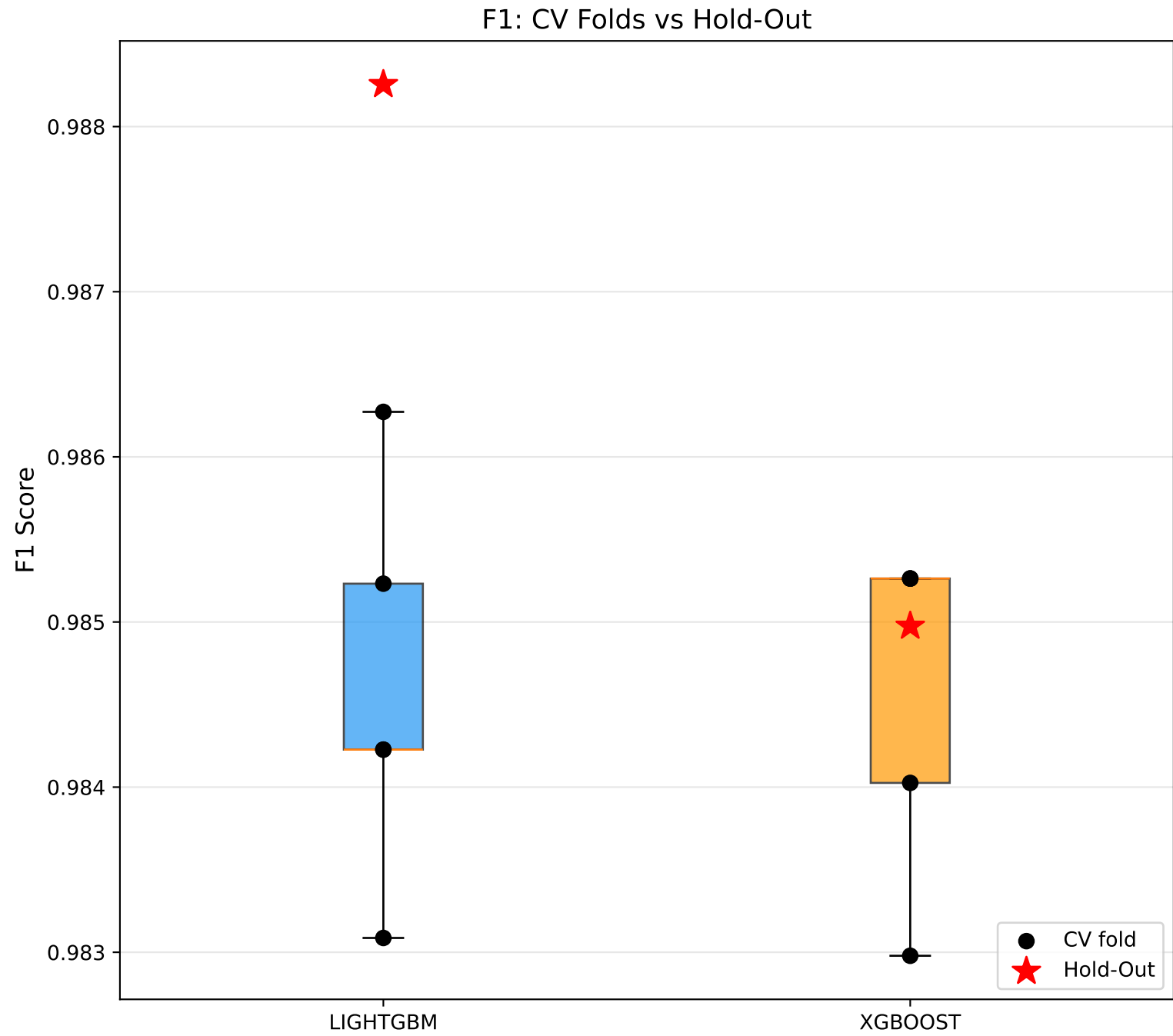
LightGBM



XGBoost



CV vs Hold-Out Karsilastirma



Detayli Sonuclar: 5-Fold CV + Hold-Out Test

Fold	Model	Split	F1	AUC-ROC	AUC-PR	Precision	Recall	MCC
1	lightgbm	CV	0.9863	0.9956	0.9983	0.9832	0.9894	0.9557
2	lightgbm	CV	0.9842	0.9959	0.9981	0.9770	0.9915	0.9489
3	lightgbm	CV	0.9842	0.9966	0.9983	0.9770	0.9915	0.9489
4	lightgbm	CV	0.9852	0.9978	0.9990	0.9811	0.9894	0.9523
5	lightgbm	CV	0.9831	0.9970	0.9986	0.9810	0.9852	0.9454
1	xgboost	CV	0.9853	0.9960	0.9984	0.9791	0.9915	0.9523
2	xgboost	CV	0.9853	0.9958	0.9981	0.9791	0.9915	0.9523
3	xgboost	CV	0.9830	0.9963	0.9982	0.9872	0.9788	0.9460
4	xgboost	CV	0.9840	0.9977	0.9990	0.9893	0.9788	0.9495
5	xgboost	CV	0.9853	0.9967	0.9985	0.9791	0.9915	0.9521
ORT	lightgbm	CV	0.9846	0.9966	0.9985	0.9799	0.9894	0.9502
ORT	xgboost	CV	0.9846	0.9965	0.9984	0.9827	0.9864	0.9505
-	lightgbm	HOLD-OUT	0.9883	0.9988	0.9995	0.9800	0.9966	0.9619
-	xgboost	HOLD-OUT	0.9850	0.9983	0.9992	0.9720	0.9983	0.9512